



1. START
GAME



2.
ACCELERATE



3. REVERSE



6. TURN
RIGHT



4. BRAKE



5. TURN LEFT



7. BALANCE

Hand Gesture Controller for Navigation Gameplay

Computer Vision-Based HCI in Gaming

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Hand Gesture Controller for Navigation Gameplay

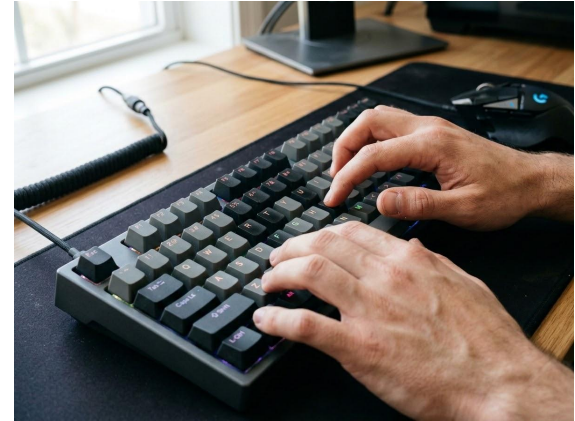
Built-in Webcam enable Natural, No Hardware Gaming

- Utilizes standard built-in laptop webcams.
- Replaces expensive motion sensors and physical gamepads.
- Provides an accessible, highly portable alternative for casual players.
- Transforms natural hand movements into intuitive game input.



Conventional Controllers Create Physical and Financial Problem

- Repetitive button pressing may causes finger fatigue and strain.
- Physical buttons act as a barrier to natural interaction.
- Motion-tracking alternatives (like VR or Kinect) are platform-specific and costly.



"Computer vision bridges this gap affordably."

— OPPORTUNITY

Constraints

Overcoming Constraints

- ✓ **Hardware**
Zero extra sensors; relies solely on a standard 720p webcam.
- ✓ **Environment**
Must handle complex backgrounds and varying lighting conditions.
- ✓ **Responsiveness**
Requires sub-150ms latency for continuous vehicle steering.
- ✓ **Accuracy**
Precise hand landmark tracking without depth-sensing hardware.

The System Architecture and Process

System Architecture

Core Technology Stack

Computer Vision

Python (OpenCV)

Node.js WebSockets

Render Engine

Phaser.js frameworks

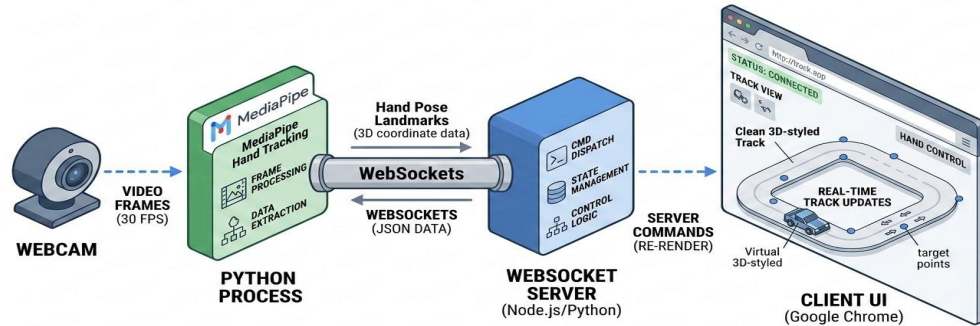


FIG 1. Pipeline flow from webcam capture to browser client

User Study: 31 participants defined intuitive gestures (e.g., open palm for 'Brake').

Dataset: 1,400 labeled images across 7 gesture classes.

ML Model: Custom-trained via MediaPipe Model Maker.

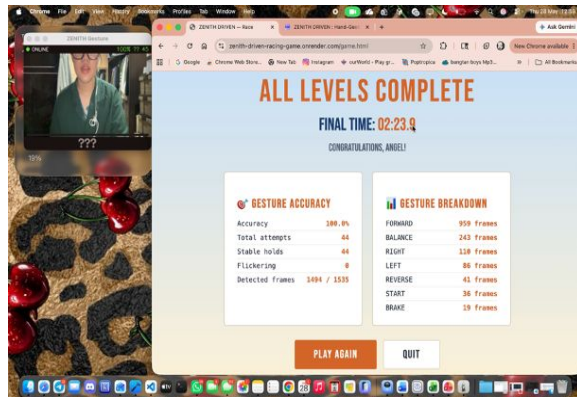
Validated Project Success

Delivering High Accuracy and Instant Usability

>90%

Accuracy

Real-time gesture
recognition accuracy.

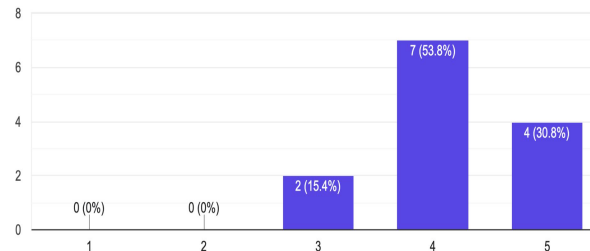


4.15/5

Satisfaction

Average rating from 13 test
participants.

Overall, how would you rate this Final Year Project?
13 responses

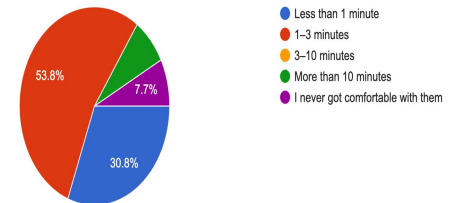


84.6%

Learning Curve

Of users mastered controls
in under 3 minutes.

How long did it take you to learn the gesture controls?
13 responses





Accessible Gaming Requires No Extra Hardware

True accessibility leverages existing everyday technology.

Key Benefits

- Built-in cameras provide a viable, low-cost input method.
- Eliminates expensive hardware barriers for inclusive, portable gaming.

Future Work

- Enhance extreme low-light stability.
- Expand gesture mapping capability.